Hackathon Problem 2: Hypertension, Explained

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Problem 2 asks for a tool that helps detect hypertension early and follow patients continuously between consultations, while the doctor keeps every medical decision.

The problem, in plain English

Hypertension is common, often silent, and poorly followed between consultations, so dangerous changes are seen too late.

What the organisers tell us

1. **It is a major public health issue.** Hypertension (high blood pressure) affects a large part of the population.
2. **It can stay undetected for a long time.** Most people feel nothing, even with very high pressure, until a stroke, heart attack or kidney damage. This is why it is called a silent disease.
3. **It is often not controlled well enough.** Even diagnosed patients on treatment often keep a pressure above target.
4. **Follow-up relies on one-off measurements.** The doctor measures the pressure only during consultations, every few weeks or months.
5. **So the doctor cannot see what happens between two visits.** Blood pressure changes during the day and over weeks (stress, salt, forgotten pills, a new treatment). One reading in the office does not show this evolution and can be misleading: some patients are higher at the doctor's, others are higher at home.
6. **High-risk patients need it most.** The brief does not define them; typically older people, diabetics, or people with heart or kidney disease.

The three goals they name

- **Early detection** ("détection précoce"): find people with high pressure sooner.
- **Regular follow-up** ("suivi régulier"): see the pressure continuously, not only at visits.
- **Identify situations that need medical action** ("identification des situations nécessitant une intervention médicale"): tell the doctor when a patient needs them.

The figures they give

- About **7.7 million people** affected in Algeria, according to their initial analysis.
- About **6.4 million** of them have **uncontrolled** hypertension: their pressure stays too high despite diagnosis or treatment.
- The theme scored **74.8 / 100** in their comparison, the second best of the problems they analysed.

The 6.4 million figure is the key one: the main gap is not only diagnosis, it is the lack of follow-up that would show a treatment is not working.

Key French terms

French in the brief	Meaning
Détection précoce	Early detection: finding hypertension before complications
Suivi continu / suivi régulier	Continuous / regular follow-up between consultations
Mesures ponctuelles	One-off measurements, taken only at the doctor's
Tensiomètre	Blood pressure monitor
Tension artérielle / pression artérielle	Blood pressure
Hypertension non contrôlée	Uncontrolled hypertension: pressure stays above target
Valeurs inhabituelles	Unusual values: a single abnormal reading
Tendances préoccupantes	Worrying trends: a slow drift over time
Professionnel de santé	Health professional (doctor, possibly nurse or pharmacist)
Aide à la décision	Decision support: the tool informs, the human decides
Dossiers médicaux électroniques	Electronic medical records (hospital systems)
Communications à faible débit	Low-bandwidth communication: SMS, weak networks

French in the brief	Meaning
Indicateurs mesurables	Measurable indicators: numbers that prove impact
Jugement clinique	Clinical judgement: the doctor's own medical decision

What they ask for

The brief suggests one shape of solution: an affordable blood pressure monitor connected to a follow-up platform that collects readings and alerts the health professional.

#	Function in the brief	What it means concretely
1	Associate an accessible blood pressure monitor ("tensiomètre accessible")	A cheap, common home monitor; no expensive or special device
2	A follow-up platform that **centralises the measurements**	Every reading, wherever it is taken, lands in one place linked to the patient
3	**Identify unusual values**	Flag a single abnormal reading (very high or very low)
4	**Identify worrying trends**	Flag a slow drift over days or weeks, even when no single reading is alarming
5	**Generate alerts for the health professional**	Notify the doctor about patients who need attention
6	Support early detection	Help find people who have hypertension but do not know it
7	Support regular follow-up	Readings between visits, not only at the doctor's
8	Focus on high-risk patients	Closer follow-up for patients most likely to have complications

The rule they state clearly

> "La décision médicale resterait sous la responsabilité du professionnel, la technologie intervenant comme outil d'aide au suivi."

In English: the medical decision stays with the professional; the technology only helps follow-up. A solution may inform, sort, explain and suggest, but never diagnose, prescribe or change a treatment on its own.

Why they chose it: their criteria, decoded

The "why this problem was chosen" list shows what the organisers value, so the jury will likely judge solutions on the same points.

What they wrote	What it really means	What a solution needs
Score 74.8 / 100, second best problem	They see it as important and realistic	A credible, complete prototype
7.7 million affected, about 6.4 million uncontrolled	The problem is huge and mostly about follow-up, not only diagnosis	Strong continuous follow-up that catches uncontrolled patients
Infrastructure compatibility 9/10, simple technologies, low-bandwidth communication	It must run on what Algeria already has	Works with cheap monitors, basic phones and SMS, not only smartphones and 4G
Low dependence on electronic medical records	Many clinics have no digital records system	A standalone platform; no hospital integration required
Geographic accessibility, urban and rural	Rural patients must be included	Access for people far from doctors and without internet
Limited extra workload for professionals, automated alerts	Doctors are already overloaded	Prioritised, automatic alerts instead of raw data to sort

What they wrote	What it really means	What a solution needs
Measurable indicators: measurement frequency, value evolution, number of alerts, follow-up regularity	They want proof of impact with numbers	These metrics, tracked and shown
Decision-support approach	The tool helps, the doctor judges	Explanations and suggestions only; no automatic medical decisions

The four indicators they name are a direct checklist: **measurement frequency, evolution of values, number of alerts, regularity of follow-up**.

What they expect as a result

They expect a working prototype that shows the full loop: a patient measures at home, the readings are centralised, the system detects a problem, and the doctor is alerted and acts.

Expected by the end of the hackathon

1. **A way for patients to record readings** taken at home with a simple monitor, easy enough for elderly users.
2. **A central platform** that stores every reading per patient and shows its evolution over time.
3. **Automatic detection** of both unusual single values and worrying trends.
4. **Alerts for the health professional**, filtered and prioritised so they do not add workload.
5. **A professional's view** to follow patients and see who needs attention.
6. **Measurable indicators** that prove the system works.
7. **A clear demo** of a realistic scenario, plus a pitch that explains impact and feasibility in Algeria.

Out of scope, or to avoid

- Replacing the doctor, diagnosing, or changing treatment automatically.
- Requiring integration with hospital information systems or electronic medical records.

- Requiring expensive connected devices or a smartphone for every patient.
- Flooding the doctor with notifications for every reading.
- A plain blood pressure diary with no professional side and no alerts: it would miss the core of the brief.